

BIOLOGY — GRADE 11 | CELL DIVISION

Quick Check

Lesson 5: Significance of Mitosis and Meiosis: Case Studies from Growth, Healing, Roots and Offspring

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.4: Cell Division | Lesson 5

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A boy is cut by a panga while weeding the shamba. After two weeks the skin has healed. Which statement best explains how the wound healed?

- A** The cells beside the cut divided by mitosis, making identical new skin cells to replace the damaged ones
- B** Meiosis made new skin cells with half the chromosomes to fill the cut
- C** Meiosis made four different skin cells from each damaged cell
- D** No cell division happened; the old skin cells just stretched over the cut

A farmer cuts a piece of sweet potato vine and plants it. After some weeks it has grown new roots and shoots. Which type of cell division produced the new roots and shoots, and what is the evidence?

- A** Mitosis, because the new cells are identical to the parent cells and have the same chromosome number
- B** Meiosis, because the new parts are made from gametes
- C** Meiosis, because plants need variation to grow roots
- D** Mitosis, because it halves the chromosome number in each new cell

Two brothers in the same family differ in height and skin tone, even though they have the same parents. Which explanation uses the correct cell division and evidence?

- A** Mitosis in the skin cells of each brother changes his height and skin tone
- B** Meiosis makes gametes that are genetically different from each other, so children differ from each other and from their parents
- C** Mitosis makes gametes that are identical, so children copy one parent exactly
- D** Meiosis makes four identical gametes with a full set of chromosomes, so the children differ

A human body cell has 46 chromosomes. Suppose gametes were made by mitosis instead of meiosis, so each sperm and each egg had 46 chromosomes. How many chromosomes would the fertilised egg have?

A 92

B 69

C 23

D 46

A baby of 3 kg grows into a teenager of 50 kg. Which explanation is best supported by the evidence from the case studies?

- A** Food turns directly into new cells without any cell division
- B** The body cells only get bigger, and no new cells are made
- C** Mitosis makes many new identical cells using materials from food, and each new cell has the full set of chromosomes (instructions)
- D** Meiosis in body cells makes the extra cells needed for growth

Which statement correctly compares the cells made by mitosis and by meiosis?

- A** Mitosis makes four different haploid cells; meiosis makes two identical diploid cells
- B** Mitosis makes gametes for reproduction; meiosis makes skin cells for healing
- C** Both make two identical cells, but meiosis is used only in plants
- D** Mitosis makes two identical diploid cells for growth and repair; meiosis makes four different haploid gametes

On a class model, a parent cell with 46 chromosomes divides to make sperm cells with 23 chromosomes each. Which label and purpose should be written on the arrow?

- A** Mitosis: to make identical cells for growth
- B** Meiosis: to make gametes with half the chromosomes, so fertilisation restores 46 and the offspring show variation
- C** Meiosis: to make body cells with 46 chromosomes for growth
- D** Mitosis: to repair damaged tissue

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).